Lesson 2.2

Conditional Statements

If... then... — and its converse, inverse and contrapositive

If p, then q

If it rains, then the ground is wet.

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p = **hypothesis** (the "if" part) q = **conclusion** (the "then" part) written $p \to q$

Four related statements

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Converse $q \to p$ **switch**: If the ground is wet, then it rained.

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Four related statements

Conditional $p \to q$ If it rains, then the ground is wet.

Converse $q \to p$ **switch**: If the ground is wet, then it rained.

Inverse $\sim p \to \sim q$ **negate**: If it does not rain, then the ground is not wet.

Contrapositive $\sim q \to \sim p$ **switch and negate**: If the ground is not wet, then it did not rain.

True or false?

Conditional: If it rains, the ground is wet. **true**

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Converse: If the ground is wet, it rained. **false** — a sprinkler!

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Converse: If the ground is wet, it rained. **false** — a sprinkler!

Contrapositive: If the ground is not wet, it did not rain. **true**

The rule

A conditional and its **contrapositive** are always both true or both false.
The **converse** and the **inverse** also match each other — but they may not match the original.

Biconditional

If a conditional **and** its converse are both true, write "**if and only if**".

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If a conditional **and** its converse are both true, write "**if and only if**".

An angle is right **if and only if** it measures 90°.

Your turn! Pause the video and try it.

"If an angle measures $30°$, then it is acute."
Write the **converse**. Is it true?

Your turn! Pause the video and try it.

"If an angle measures 30°, then it is acute."
Write the **converse**. Is it true?

Converse: If an angle is acute, then it measures 30°. false
counterexample: a 50° angle is acute.

Remember

- Conditional $p \to q$: **converse** switches, **inverse** negates, **contrapositive** does both.
- A conditional and its contrapositive are both true or both false.
- Both directions true? Use "**if and only if**".

Now practice at your level:

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